



FEASIBILITY STUDY
LITPATH AI: SMART PATHFINDER FOR THESES AND DISSERTATIONS

ASIA PACIFIC COLLEGE
3 HUMABON, MAGALLANES
MAKATI CITY, METRO MANILA 1232

APRIL 2026





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1. EXECUTIVE SUMMARY

The executive summary provides an overview of the content contained in the feasibility study document. Many people write this section after the rest of the document is completed. This section is important in that it provides a higher level summary of the detail contained within the rest of the document.

The DOST-Science and Technology Information Institute (DOST-STII) Library maintains a significant collection of graduate theses and dissertations produced by DOST Scholars, cataloged through its Online Public Access Catalog (OPAC). Despite the academic value of this collection, it remains consistently underutilized. The existing OPAC relies on exact keyword-based searches, suffers from processing times exceeding one minute per query, and lacks any artificial intelligence capability. These shortcomings create a critical barrier to knowledge discovery, preventing students, researchers, and DOST personnel from efficiently accessing and building upon existing research.

LitPath AI is a proposed standalone web application developed by HecTech, a four-member team from Asia Pacific College, to address these challenges directly. The system introduces an AI-powered semantic search engine built on a Retrieval-Augmented Generation (RAG) pipeline using SentenceTransformers for embedding generation, ChromaDB for vector storage, and Gemini AI for query understanding and result summarization. Additional features include an automated multi-format citation generator, research bookmarking, a user feedback mechanism, and an administrative analytics dashboard for DOST-STII library staff. The platform will be deployed as a publicly accessible web application via Cloudflare, developed using a React.js frontend and a Django backend.

This feasibility study assesses the viability of the LitPath AI project across technical, organizational, market, and financial dimensions. Findings indicate that the project is technically achievable using the defined open-source and cloud technology stack, operationally manageable within the existing DOST-STII structure, and financially sound within the scope of an academic project engagement. The primary open cost items, AI model API access and Cloudflare deployment tier, are currently under evaluation by the team and are noted in the financial projections as pending finalization. The study recommends proceeding with project approval and initiation of formal development.

2. DESCRIPTION OF PRODUCTS AND SERVICES

This section provides a high level description of the products and/or services which are being considered as part of the feasibility study. The purpose of this section is to provide detailed descriptions of exactly what the organization is considering so this information can be applied to the following sections of the document. It is important that this description captures the most important aspects of the products and/or services that the organization is considering as well as how it may benefit customers and the organization.



LitPath AI is an AI-powered search engine platform developed exclusively for the DOST-STII Library. The system is designed to dramatically improve how students, academic researchers, and DOST employees search for and interact with the library's collection of graduate theses and dissertations. Unlike the current OPAC, LitPath AI employs a semantic search engine backed by a Retrieval-Augmented Generation (RAG) pipeline to interpret the meaning and research intent behind user queries, surfacing relevant results even when users are unfamiliar with precise academic terminology.

The platform is delivered as a fully functional standalone web application, accessible via cloud deployment on Cloudflare so that users may access it from any location without software installation. The system operates on a backend database populated exclusively with thesis and dissertation metadata and abstracts provided directly by DOST-STII. No changes to the existing OPAC infrastructure are required. The core features of LitPath AI are as follows:

1. **AI-Powered Semantic Search Engine:** A search module that accepts natural language queries, converts them into vector embeddings via SentenceTransformers, and retrieves semantically relevant results from a ChromaDB vector database. Results are reranked using AI to maximize relevance. The system targets a minimum relevance accuracy of 70% based on user feedback.
2. **AI-Generated Overview Summaries:** Gemini AI generates contextual overview summaries of search results with embedded citations, enabling users to quickly assess relevance without reading full documents.
3. **Automated Citation Generator:** A built-in tool that automatically formats citations for selected materials in APA, MLA, Chicago Manual of Style, and IEEE formats, eliminating manual citation compilation.
4. **Research Bookmarking and History Tracking:** Users can save and organize discovered research materials and revisit their search history within their session.
5. **User Feedback Tool:** An integrated mechanism allowing users to rate search results and submit comments, providing continuous data for system validation and improvement.
6. **Admin Analytics Dashboard:** A dedicated interface for DOST-STII library administrators to monitor search trends, usage patterns, and user engagement, supporting data-driven library management.

3. TECHNOLOGY CONSIDERATIONS

This section should explain any considerations the organization must make with regards to technology. Many new initiatives rely on technology to manage or monitor various business functions. New technology may be developed internally or contracted through a service provider and always result in costs which must be weighed in determining the path forward.



LitPath AI is built on a well-defined, modular technology stack selected to balance capability, cost, and implementation risk within the constraints of a four-member academic development team. The architecture follows a three-tier model: a Presentation Layer (frontend), an Application Layer (backend and AI processing), and a Data Layer (relational and vector databases). Each component is described below.

Frontend

- React.js for a dynamic, component-based user interface.
- Tailwind CSS for responsive, utility-first styling.
- Vite as the build tool for fast development and optimized production builds.

Backend

- Django (Python) as the web framework, exposing a RESTful API architecture to the frontend.
- Handles user authentication, session management, bookmarking, citation generation, feedback collection, and admin analytics logic.

AI and Semantic Search Components

- SentenceTransformers: Generates vector embeddings from user queries and thesis metadata, enabling semantic similarity matching beyond keyword lookup.
- ChromaDB: A vector database that stores document embeddings and supports fast, high-dimensional similarity search for ranked result retrieval.
- Gemini AI API: Powers query understanding, result reranking, and AI-generated overview summaries with citations. The team is currently evaluating which tier of the Gemini API (free vs. paid) is sustainable for production use. This cost item is pending finalization.
- Retrieval-Augmented Generation (RAG) Pipeline: Combines embedding-based retrieval with Gemini AI generation to produce contextually grounded summaries and search results.

Database Systems

- Supabase (PostgreSQL): Relational database storing user accounts, sessions, bookmarks, search history, feedback data, and activity logs.
- ChromaDB: Vector database storing semantic embeddings of thesis documents. The hybrid architecture ensures both structured data queries and fast semantic retrieval are supported.

Deployment and Infrastructure



- Development and Testing: Currently performed on a local Proxmox-based server (HP ProDesk 600 G2 MT, Intel Core i5-6500, 32 GB RAM, 512 GB SSD), which is sufficient for development, testing, and beta deployment phases.
- Production Deployment: The team is planning cloud deployment via Cloudflare, which provides global CDN distribution, DDoS protection, and SSL/TLS termination. The appropriate Cloudflare service tier (free vs. paid Workers/Pages plan) is currently under evaluation. This cost item is pending finalization.
- Security: The system implements HTTPS communication, password hashing and encryption, role-based access control (User and Admin roles), secure token-based session management, firewall and network restrictions, and logging of user activities for audit and monitoring purposes.

The use of open-source frameworks (React, Django, SentenceTransformers, ChromaDB) eliminates licensing costs for most of the stack, substantially reducing the project's cost base. The two remaining open cost items, Gemini AI API access and Cloudflare deployment tier, are the only external expenses requiring resolution before final budget confirmation. Both are commercially available services with free tiers that may be sufficient for an initial deployment phase, with upgrade paths available should usage scale beyond free-tier limits.

4. PRODUCT/SERVICE MARKETPLACE

This section describes the existing marketplace for the products and/or services the organization is considering. It may describe who the target market consists of for these products or services, who the competitors are, how products will be distributed, and why customers might choose to buy our products/services. Most marketplaces are dynamic environments in which things change constantly. To enter a new marketplace blindly will usually result in an organization not fully understanding its role and not maximizing its resulting benefits.

The market for AI-powered academic research tools is experiencing rapid growth globally, driven by increasing demand for efficient knowledge discovery in higher education and research institutions. Within the Philippine government science sector, this trend is especially relevant as DOST actively invests in digital transformation initiatives that improve access to its knowledge assets.

LitPath AI serves two distinct user roles. The primary users are students and academic researchers engaged in thesis and dissertation research, as well as DOST employees conducting literature reviews and knowledge discovery activities. The secondary users are system administrators, divided into two roles: IT Administrators, who are responsible for managing user accounts, configuring AI model parameters and APIs, maintaining database structure and backups, overseeing system security, integrating LitPath AI with other



institutional systems, monitoring performance and uptime, and deploying updates and new features; and Librarian Administrators, who review and validate user feedback, generate usage reports, and review search results for accuracy.

The current landscape of competing tools and alternatives that users rely on includes:

- Google Scholar: The dominant free academic search tool globally. While highly capable of general academic literature, it does not index the DOST-STII library's internal thesis collection, meaning DOST Scholars' work remains undiscoverable through this platform.
- University and Institutional OPAC Systems: Philippine universities and research institutions operate OPAC systems for internal catalog management. These systems, including DOST-STII's own OPAC, share the same fundamental limitation of exact keyword matching and lack semantic or AI-powered search capability.
- ResearchGate and Scopus/Web of Science: These platforms serve the broader global research community but require institutional subscriptions for full access and do not include unpublished theses or grey literature from DOST-STII's collection.

LitPath AI's key competitive differentiator is its exclusive, purpose-built focus on the DOST-STII thesis and dissertation collection combined with a RAG-based AI search layer. No existing commercial or open-access platform provides semantic search specifically over this dataset. This positions LitPath AI not as a competitor to Google Scholar or Scopus, but as a complementary, institution-specific discovery tool that unlocks value from a previously underutilized collection. Distribution of the platform occurs through cloud-based web access via Cloudflare, requiring no software installation from users.

5. MARKETING STRATEGY

This section provides a high level description of how the organization will market its product or service. Some topics which should be included are: how does an organization differentiate itself from its competitors; types of marketing the organization will utilize; and who the organization will target. Marketing efforts must be focused on the right target groups in order to yield the greatest return on investment.

As a government-funded academic tool, LitPath AI will not engage in commercial advertising. Its adoption strategy focuses on institutional promotion, structured user onboarding, and demonstrating value through measurable performance improvements. The goal is to maximize awareness and active usage among the target user groups: DOST Scholars, students, academic researchers, and DOST personnel.



- **Institutional Endorsement and Communication:** DOST-STII will promote LitPath AI through official internal communications, including email circulars, the DOST-STII website, and announcements distributed to partner universities and DOST scholarship recipients. The institutional credibility of DOST-STII serves as a strong endorsement for the platform.
- **Differentiation Through Relevance and Speed:** LitPath AI's primary differentiating value proposition is its ability to display relevant research faster and more accurately than any existing tool available to DOST Scholars. Performance benchmarks comparing LitPath AI directly against the current OPAC, in terms of query response time and result relevance, will be used in onboarding materials and institutional communications to demonstrate concrete improvement.
- **User Onboarding and Documentation:** A comprehensive user guide will be made available within the platform and through DOST-STII library staff. Administrators will be trained to assist users during reference interviews, ensuring smooth adoption.
- **Feedback-Driven Improvement and Credibility:** The integrated user feedback tool will collect ratings and comments from day one of launch. Aggregated satisfaction data will be used to validate system performance against the target score of at least 3.5 out of 5 and will be shared with stakeholders to build confidence in the system.
- **Beta Testing as Social Proof:** A structured beta testing phase involving DOST employees and students will precede full launch. Positive feedback gathered during this phase will be used to refine the system and support broader institutional promotion.

Because the platform is free to all authorized users, there are no subscription or pricing barriers to adoption. Marketing costs are minimal, focused primarily on documentation preparation and internal communications. The primary success metric for reach will be active user volume and average satisfaction scores.

6. ORGANIZATION AND STAFFING

With many new products or services there may be a need for additional staffing or for an organization to restructure in order to accommodate the change. These are important considerations as they may result in increased costs or require an organization to change its practices and processes.

The LitPath AI project is developed under the supervision of DOST-STII and executed by HecTech, a four-member development team from Asia Pacific College. The organizational



structure is lean and requires no new permanent positions or departmental changes within DOST-STII. The project operates within an existing academic and institutional framework, with clear role description between the client organization and the development team.

Client Organization: DOST-STII

DOST-STII serves as the project owner and primary stakeholder. The institute's library staff and designated project supervisors are responsible for providing the thesis and dissertation dataset (metadata and abstracts), validating system requirements, participating in user acceptance testing, and formally signing off on all deliverables. The admin analytics dashboard is specifically designed for DOST-STII library staff and requires no technical expertise to operate.

Development Team: HecTech

The HecTech team is collectively responsible for all design, development, testing, and documentation work packages. Communication is conducted through daily interactions, weekly team meetings, and sprint board updates managed in OpenProject. Each member maintains current task statuses and escalates blockers immediately to the Project Manager.

Name	Role	Technical Focus	Contact
Tracie Tomon	Project Manager	Project oversight, sprint planning, stakeholder communication	<i>tctomon@student.apc.edu.ph</i>
Charijoy Cempron	System Architect	AI/backend architecture, RAG pipeline, ChromaDB integration	<i>ctcempron@student.apc.edu.ph</i>
Jenine Elaine Dulay	UI/UX Designer	React.js frontend, Tailwind CSS, user experience flow	<i>jcdulay@student.apc.edu.ph</i>
Joshua Sta Ana	Security Lead	Django backend, authentication, admin dashboard, QA	<i>jtstaana2@student.apc.edu.ph</i>

No additional permanent hires are anticipated. Post-deployment, system operation and maintenance will be handled by DOST-STII's existing technical staff using the technical manual and documentation produced as project deliverables.



7. SCHEDULE

This section is intended to provide a high level framework for implementation of the product or service being considered. This section is not intended to include a detailed schedule as this would be developed during project planning should this initiative be approved. This section may include some targeted milestones and timeframes for completion as a guideline only.

The LitPath AI project targets completion by June-July 2026, following an agile sprint methodology with progress tracked through OpenProject. The following high-level milestones govern the project schedule:

Milestone	Target Date	Status
Project Scoping	April 2025	Completed
Information Gathering	April 2025	Completed
Completion of Project Deliverables & Documentation	June 2025	Completed
Design Thinking Process (Stages 1–5)	Aug 2025	Completed
Core UI Development (System UI, Citation Generator, Bookmark)	Sep – Oct 2025	Completed
AI Search Engine, Generate Overview, Feedback Mechanism & Admin Dashboard	Oct 2025 – Jan 2026	Completed
Feature Refinement & Client Feedback Integration	Feb – Mar 2026	Completed
Continued Development & Performance Optimization (incl. codebase restructuring)	Mar – Apr 2026	In Progress
Proxmox Server Deployment & Internal Testing	Apr 2026	In Progress
Bug Fixes & Usability Testing	May – Jun 2026	Not Started
Final Deployment	Jun – Jul 2026	Not Started

A detailed work breakdown structure and sprint schedule will be established by the HecTech team in OpenProject upon formal project approval. All milestones are subject to stakeholder review and sign-off before proceeding to subsequent phases.



8. FINANCIAL PROJECTIONS

This section provides a description of the financial projections the new initiative is expected to yield versus additional costs. Financial projections are one key aspect of new project selection criteria. There are many ways to present these projections. Net present value (NPV), cost-benefit calculations, and balance sheets are just some examples of how financial projections may be illustrated. This section should also provide the assumptions on which the illustrated financial projections are based.

The financial projections for LitPath AI reflect the cost structure of an academic technology project developed entirely through free, open-source, and zero-cost resources. As a non-commercial platform built for a government institution, no development budget was required from either DOST-STII or the HecTech team. The financial analysis distinguishes between the development phase, which incurred zero cost, and the post-handover operational phase, which will be assumed by DOST-STII as the platform owner.

The following assumptions underline these projections:

- The entire development phase was completed at zero direct cost. All tools, frameworks, libraries, and infrastructure used during development are open-source or available under free-tier plans.
- No honoraria, allowances, or personnel costs were incurred. The HecTech team completed all design, development, testing, and documentation as part of an academic project engagement.
- Development and testing were conducted using existing hardware (Proxmox-based local server), requiring no hardware acquisition.
- Free-tier plans for the Gemini AI API and Cloudflare were used throughout the development and testing phases without incurring charges.
- Upon formal project handover and deployment to production, DOST-STII will assume ownership and management of all operational costs, specifically the Gemini AI API subscription and Cloudflare hosting plan, scaled to actual production usage.
- Post-launch system maintenance will be handled by DOST-STII's existing IT Administrator and Librarian Administrator roles, requiring no additional staffing costs.

Cost Item	Dev Phase (HecTech)	Post-Handover (DOST-STII)
DEVELOPMENT COSTS (HecTech)		
Personnel, Honoraria & Allowances	₱0	—



Software Licenses & Tools (React, Django, SentenceTransformers, ChromaDB, Supabase)	₱0	—
Hardware (Existing Proxmox server — no new acquisition)	₱0	—
Testing, QA, & Documentation	₱0	—
Gemini AI API (free tier used during development)	₱0	—
Cloudflare Deployment (free tier used during development)	₱0	—
TOTAL DEVELOPMENT COST (HecTech)	₱0	—
POST-HANDOVER OPERATIONAL COSTS (DOST-STII)		
Gemini AI API (production tier — scaled to usage volume)	—	To be determined by DOST-STII
Cloudflare Hosting Plan (production tier — scaled to traffic)	—	To be determined by DOST-STII
System Maintenance (existing IT Admin & Librarian Admin roles)	—	₱0 (existing staff)

Note: System maintenance is handled by DOST-STII's existing IT Administrator and Librarian Administrator staff at no additional personnel cost.

The project delivers a fully operational AI-powered research platform at zero development cost to the institution. The only post-handover expenses are the Gemini AI API and Cloudflare hosting, both of which offer scalable pricing tiers, including free tiers that may remain sufficient depending on production traffic volume. DOST-STII's IT Administrator will evaluate and select the appropriate service tiers upon taking ownership of the system. The institutional value delivered, a measurably faster and more efficient research experience compared to the existing OPAC, improved search relevance to a minimum of 70%, and a modern AI-powered discovery platform for DOST Scholars and researchers, significantly outweighs any ongoing operational costs.



9. FINDINGS AND RECOMMENDATIONS

This section should summarize the findings of the feasibility study and explain why this course of action is or is not recommended. This section may include a description of pros and cons for the initiative being considered. This section should be brief since most of the detail is included elsewhere in the document. Additionally, it should capture the likelihood of success for the business idea being studied.

Based on the analysis presented across all sections of this feasibility study, it is recommended that DOST-STII formally approve the LitPath AI project and authorize the HecTech team to proceed with development. The study findings consistently support the technical viability, organizational readiness, market relevance, and financial soundness of the proposed system. The only open items, Gemini AI API tier and Cloudflare deployment plan, are minor decisions with low-cost or free options available, and do not represent blockers to project initiation.

Key findings by dimension are summarized below:

Technology

- The full technology stack (React.js, Django, SentenceTransformers, ChromaDB, Supabase, RAG pipeline) is defined, proven, and in use in comparable academic systems globally, substantially reducing technical risk.
- Open-source frameworks eliminate licensing costs for most of the stack.
- Existing Proxmox-based hardware is sufficient for development and testing phases, avoiding hardware capital expenditure.
- Cloudflare deployment provides enterprise-grade CDN, security, and global access at potentially zero cost using its free tier.
- The HecTech team has technical competencies, covering system architecture, AI/ML, frontend design, backend development, and security, to deliver the full system within the June 2026 deadline.

Market and User Need

- A clear and documented gap exists between current OPAC capability and user needs: fewer than 70% of results match user intent, queries take over a minute, and essential features like citation generation are absent.
- No existing platform provides semantic search specifically over the DOST-STII thesis collection, confirming that LitPath AI fills a genuine, unmet institutional need.
- The target user base is well-defined and accessible through DOST-STII's existing communication channels.



Organization

- No new permanent staffing or structural changes are required within DOST-STII.
- Post-deployment maintenance can be handled by DOST-STII's existing technical staff using the project's documentation.
- The HecTech team's structured use of OpenProject for sprint tracking and daily communication ensures accountability and on-time delivery.

Financial

- The entire development phase was completed at zero cost to both HecTech and DOST-STII. All tools, frameworks, hardware, and cloud services used during development were sourced from free and open-source solutions.
- Post-handover, the only operational expenses are the Gemini AI API and Cloudflare hosting, both of which will be managed and funded by DOST-STII's IT Administrator. Both services offer free tiers that may remain sufficient depending on production usage volume.
- The absence of licensing fees, personnel costs, and hardware acquisition makes LitPath AI an exceptionally cost-efficient solution for DOST-STII.
- The project delivers a fully functional, production-ready AI platform at no direct cost to the institution during development, a strong return on institutional investment.

In conclusion, LitPath AI represents a high-value, zero-cost investment for DOST-STII during the development phase. The project directly addresses a well-documented institutional problem with a well-defined technical solution, a capable development team, and a clear June-July 2026 delivery target. With the entire development completed at no cost and post-handover operational expenses remaining minimal and manageable, approval of this project is strongly recommended.